2025.11.02 Deep learning I

Exercise 3 optimization I

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Exercise Sheet 3

Exercise 1: Neural Network Optimization (20 + 20 + 15 P)

Consider the one-layer neural network

$$f(\boldsymbol{x}) = \boldsymbol{w}^\top \boldsymbol{x}$$

applied to data points $\boldsymbol{x} \in \mathbb{R}^d$, and where $\boldsymbol{w} \in \mathbb{R}^d$ is the parameter of the model. We would like to optimize the mean square error objective:

$$J(\boldsymbol{w}) = \mathbb{E}_{\hat{p}}\Big[\frac{1}{2}(\boldsymbol{w}^\top \boldsymbol{x} - t)^2\Big],$$

where the expectation is computed over an empirical approximation $\hat{p}$ of the true joint distribution $p(\boldsymbol{x}, t)$. The ground truth is known to be of type: $t|\boldsymbol{x} = \boldsymbol{v}^\top \boldsymbol{x} + \varepsilon$, with the parameter $\boldsymbol{v}$ unknown, and where ε is some small i.i.d. Gaussian noise. The input data follows the distribution $\boldsymbol{x} \sim \mathcal{N}(\boldsymbol{\mu}, \sigma^2 I)$ where $\boldsymbol{\mu}$ and σ^2 are the mean and variance.

(a) *Compute* the Hessian of the objective function J at the current location $\boldsymbol{w}$ in the parameter space, and as a function of the parameters $\boldsymbol{\mu}$ and σ of the data.

(b) *Show* that the condition number of the Hessian is given by: $\dfrac{\lambda_1}{\lambda_d} = 1 + \dfrac{\|\boldsymbol{\mu}\|^2}{\sigma^2}$.

(c) *Explain* for this particular problem what would be the advantages and disadvantages of centering the data before training. You answer could include the following aspects: (1) condition number and speed of convergence, (2) ability to reach a low prediction error.

Exercise 2: Initialization (35 + 10 P)

Consider a deep neural network with L layers with width n and a ReLU activation function. Assume the dataset X, which consists of samples $x \in \mathbb{R}^n$ which are iid.. X is centered and whitened, i.e., $\mathbb{E}[x^{(i)}] = 0$ and $\mathrm{Var}[x^{(i)}] = 1 \forall i \in \{1, \ldots, n\}$, $\mathrm{Cov}(x^{(i)}, x^{(j)}) = 0 \forall i \neq j$ where i, j indicate the dimensions.
The He-initialization is defined as follows:

$$W_{ij}^{(l)} \sim \mathcal{N}\left(0, \frac{2}{n}\right)$$
$$b_i^{(l)} = 0\,,$$

where $W_{(l)}$ is the weight matrix of layer l and $b^{(l)}$ is the bias vector of layer l.
You may use the following assumptions/hints:

- For a random variable Y centered around 0, i.e. $\mathbb{E}[Y] = 0$, we assume $\mathbb{E}[\mathrm{ReLU}(Y)^2] = \frac{1}{2}\mathrm{Var}(Y)$.
- For mutually independent random variables a, b, we have $\mathbb{E}[ab] = \mathbb{E}[a]\mathbb{E}[b]$.
- $\mathbb{E}[\sum_i Y^{(i)}] = \sum_i \mathbb{E}[Y^{(i)}] = n\mathbb{E}[Y]$.
- a_0 is the input to the neural network.

(a) *Show* by induction that, when using the initialization scheme of He et al. (2015), the variance of the latent variables $z_{(l)}^{(i)} = \sum_j W_{(l)}^{(ij)} a_{(l-1)}^{(j)} + b_{(l)}^{(i)}$ for all layers $l \in \{2, ..., L\}$ stays constant, i.e. $\mathrm{Var}(\sum_j W_{(l+1)}^{(ij)} a_{(l)}^{(i)} + b_{(l+1)}^{(i)}) = \mathrm{Var}(\sum_j W_{(l)}^{(ij)} a_{(l-1)}^{(j)} + b_{(l)}^{(i)})$.

(b) How do you need to choose the initialization for tanh if you assume the following?

- Around 0, we have $\tanh(x) \approx x$.

Show your work.

Exercise 1

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(a) *Compute* the Hessian of the objective function J at the current location $\boldsymbol{w}$ in the parameter space, and as a function of the parameters $\boldsymbol{\mu}$ and σ of the data.

$t|x = v^T x + \varepsilon$

$J(w) = \mathbb{E}_{\hat{p}}[\frac{1}{2}(w^T x - t)^2]$

$= \mathbb{E}_{\hat{p}}[\frac{1}{2}(w^T x - v^T x - \varepsilon)^2]$

$\frac{\partial J(w)}{\partial w_i} = \mathbb{E}_{\hat{p}}[\frac{1}{2}\cdot 2(w^T x - v^T x - \varepsilon)\cdot x_i] = \mathbb{E}_{\hat{p}}[(w^T x - v^T x - \varepsilon)\cdot x_i]$

$\frac{\partial^2 J(w)}{\partial w_i \partial w_j} = \mathbb{E}_{\hat{p}}[x_i \cdot x_j] = \mathbb{E}_{\hat{p}}[x x^T]$, Because $\hat{p}$ ist true joint distribution $p(x,t)$

$x \sim \mathcal{N}(\mu, \sigma^2 I)$

σ ist stand derivation, σ^2 is variance

$\sigma^2 I$ ist Covarianz

Covarianz $= Cov(X, X) = \mathbb{E}[(X - \mathbb{E}[X])\cdot(X - \mathbb{E}[X])]$

$= \mathbb{E}_{\hat{p}}[(X-\mu)(X-\mu)^T] = \sigma^2 I$

$\mathbb{E}_{\hat{p}}[(x-\mu)(x-\mu)^T] = \sigma^2 I$

$= \mathbb{E}_{\hat{p}}[xx^T - x\mu^T - \mu x^T + \mu\mu^T]$

$= \mathbb{E}_{\hat{p}}[xx^T] - \mathbb{E}_{\hat{p}}[x]\mu^T - \mu\mathbb{E}_{\hat{p}}[x]^T + \mu\mu^T$

$= \mathbb{E}_{\hat{p}}[xx^T] - \mu\mu^T - \mu\mu^T + \mu\mu^T$

$= \mathbb{E}_{\hat{p}}[xx^T] - \mu\mu^T$.

$= H(w) - \mu\mu^T$

$\Rightarrow H(w) = \sigma^2 I + \mu\mu^T$

(b) *Show* that the condition number of the Hessian is given by: $\frac{\lambda_1}{\lambda_d} = 1 + \frac{\|\mu\|^2}{\sigma^2}$.

From 1a, we get $H(w) = \sigma^2 I + \mu\mu^T$

1 Using the Rayleigh quotient to find Eigenvales

For any unit vector v ($\|v\| = 1$):

$v^T H v = v^T(\sigma^2 I + \mu\mu^T)v = \sigma^2 v^T v + (v^T\mu)^2 = \sigma^2 + (v^T\mu)^2$

The largest eigenvalue of H can be written as:

$$\lambda_1 = \max_{\|v\|=1} v^T H v = \sigma^2 + \max_{\|v\|=1} (v^T\mu)^2$$

For any real symmetric matrix, the largest eigenvalue equals the maximum Rayleigh quotient $v^T H v$ over all unit vectors v.

2

Since v is a unit Vector

$$v^T\mu = \|\mu\| \cos\theta$$

Where θ is the angle between v and μ.

$$(v^T\mu)^2 = \|\mu\|^2 \cdot \cos^2\theta \le \|\mu\|$$

The maximum happens when v is parallel to μ

Therefore

$$\lambda_1 = \sigma^2 + \|\mu\|^2$$

3 Computing λ_{max} and λ_{min}

if μ ist parallel to μ, $\sigma^2 + \|\mu\|^2 = \lambda_{max}$

if μ ist orthogonal to μ, $(v^T \cdot \mu = 0)$ the $v^T H v = \sigma^2 = \lambda_{min}$.

4 Computing condition nunmer

$$\frac{\lambda_{max}}{\lambda_{min}} = \frac{\sigma^2 + \|\mu\|^2}{\sigma^2} = 1 + \frac{\|\mu\|^2}{\sigma^2}$$

(c) *Explain* for this particular problem what would be the advantages and disadvantages of centering the data before training. You answer could include the following aspects: (1) condition number and speed of convergence, (2) ability to reach a low prediction error.

Advanteages: of centering the data befor training

1. Get lower condition number. There fore, covergence ist faster

Disadvaneges of centering the data before traing

1. Bias in mode class

when data is centered, a linear model can only ~~be~~ to be in form

$$f(x) = w^T (x - \mathbb{E}(x))$$

It's called homogenous model

If the ground truth is like. $f(x) = v^T x$. This ground Turth can not be represented by those homogenous model. so the minimal Error might be higher.

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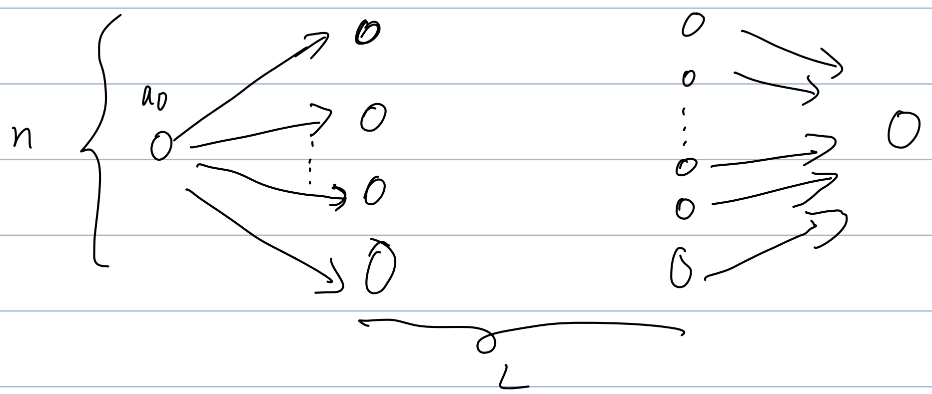


(a) *Show* by induction that, when using the initialization scheme of He et al. (2015), the variance of the latent variables $z_{(l)}^{(i)} = \sum_j W_{(l)}^{(ij)} a_{(l-1)}^{(j)} + b_{(l)}^{(i)}$ for all layers $l \in \{2, ..., L\}$ stays constant, i.e. $\mathrm{Var}(\sum_j W_{(l+1)}^{(ij)} a_{(l)}^{(i)} + b_{(l+1)}^{(i)}) = \mathrm{Var}(\sum_j W_{(l)}^{(ij)} a_{(l-1)}^{(j)} + b_{(l)}^{(i)})$.

1 $z_{(l)}^{(i)} = \sum_j W_{(l)}^{(ij)} a_{(l-1)}^{(j)} + b_{(l)}^{(i)}$

$a_{(l)}^{i} = \max(0, z_{(l)}^{(i)}) = \mathrm{ReLU}(\sum_j W_{(l)}^{(ij)} a_{(l-1)}^{(j)} + b_{(l)}^{(i)})$

$W_{(l)}^{(ij)}, a_{(l-1)}^{(j)}, b_{(l)}^{(i)}$ those there are all independence of each other.

2 For all layer $l \in (1, \ldots, L)$

$$\mathbb{E}\left[\sum_i W_{(l)}^{(ij)} a_{(l-1)}^{(i)}\right] \xlongequal[\text{independence}]{W \text{ and } a \text{ are}} \sum_i \underbrace{\mathbb{E}\left[W_{(l)}^{(ij)}\right]}_{0} \cdot \mathbb{E}\left(a_{(l-1)}^{(i)}\right)$$

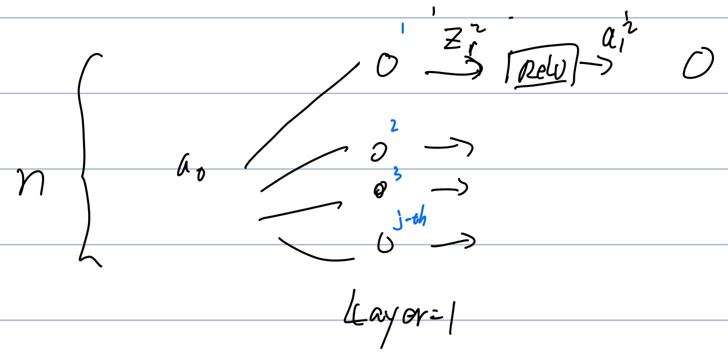
$$= 0 \cdot n \cdot \mathbb{E}\left[a_{(l-1)}^{(i)}\right] = 0$$

$$\mathbb{E}\left[\left(W_{(l)}^{(ij)}\right)^2\right] = Var\left(W_{(l)}^{(ij)}\right) + \mathbb{E}\left[W_{(l)}^{(ij)}\right]^2 \quad (\text{according to task 1a})$$

$$= Var\left(W_{(l)}^{(ij)}\right) + 0$$

3 prove the variance of the latent variables $z_{(l)}^{(j)} = \sum_i W_{(l)}^{(ij)} a_{(l-1)}^{(i)} + b_{(l)}^{(j)}$ for all layers $l \in \{1, \ldots, L\}$ stays constant.

3.1 For layer $= 1$ $(l = 1)$



$$Var(z_1^i) = Var\left(\sum_j W_{(1)}^{(ji)} a_{(0)}^{(j)} + \underbrace{b_{(0)}^{i}}_{=0}\right) = \sum_k Var\left(W_{(1)}^{(ji)} x^{(j)}\right)$$

$$= \sum_j \underbrace{Var\left(W_1^{(ji)}\right)}_{\frac{2}{n}} \cdot \underbrace{Var\left(x^{(j)}\right)}_{1}$$

$$= \sum_j \frac{2}{n} \cdot 1 = \underbrace{\left(\frac{2}{n} + \frac{2}{n} + \cdots \frac{2}{n}\right)}_{n} = 2.$$

3.2 For Layer > 1

$$Var(z_{(l)}^{(i)})$$

$$= Var\left(\sum_i W_{(l)}^{(i,j)} a_{(l-1)}^{(i)} + \underbrace{b_{(l)}^{(i)}}_{=0}\right)$$

$$= Var\left(\sum_i W_{(l)}^{(i,j)} a_{(l-1)}^{(i)}\right)$$

Them, according to the Induction in task (a)

$$= \mathbb{E}\left[\left(\sum_i W_{(l)}^{(i,j)} \cdot a_{(l-1)}^{(i)}\right)^2\right] - \mathbb{E}\left[\sum_i W_{(l)}^{(i,j)} a_{(l-1)}^{(i)}\right]^2$$

$$= \mathbb{E}\left[\left(\sum_i W_{(l)}^{(i,j)} \cdot a_{(l-1)}^{(i)}\right)^2\right] - \underbrace{\mathbb{E}\left[\sum_i W_{(l)}^{(i,j)}\right]^2}_{=0} \cdot \mathbb{E}\left[\sum_i a_{(l-1)}^{(i)}\right]^2$$

$$= \mathbb{E}\left[\sum_i W^{2\,(i,j)}_{(l)}\right] \cdot \mathbb{E}\left[\sum_i a^{2\,(i)}_{(l-1)}\right]$$

$$= \sum_i Var\left(W_{(l)}^{(i,j)}\right) \cdot \mathbb{E}\left[\sum_i a^{2\,(i)}_{(l-1)}\right]$$

$$= \sum_i Var\left(W_{(l)}^{(i,j)}\right) \cdot \mathbb{E}\left[a^{2\,(i)}_{(l-1)}\right]$$

$$= \sum_i \frac{2}{n} \mathbb{E}\left[a_{(l-1)}^{(i)}\right]$$

$$= \sum_i \frac{2}{n} \mathbb{E}\left[\left(ReLU(z_{(l-1)}^{(i)})\right)^2\right]$$

$$= \sum_i \frac{2}{n} \cdot \frac{1}{2} Var\left(z_{(l-1)}^{(i)}\right)$$

$$= \frac{1}{n} \sum_i Var\left(z_{(l-1)}^{(i)}\right)$$

$$= Var\left(z_{(l-1)}^{(i)}\right)$$

$$= 2 \quad \text{(according to the induction in task 2.3.1}$$

(b) How do you need to choose the initialization for tanh if you assume the following?

- Around 0, we have $\tanh(x) \approx x$.

1 $\tanh(x) \approx x$

(Regime Region)

around $x=0$ region, the funcetion ist almost linear.
Small change in x lead to proportional changes in $\tanh(x)$.
In this Region, the slope is roughly 1

2 With the Lineariety in Regime Region,
$\mathbb{E}[\tanh(z^{(i)}_{(l)})^2]$ is approximated to $\mathbb{E}[(z^{(i)}_{(l)})^2]$.

3 In order to achieve that,
According to Inductiona in task 2 (2.3.2)

$$Var(z^{(i)}_{(l)})$$
$$= \cdots$$
$$= \sum_i Var(w^{(ij)}_{(l)}) \cdot \mathbb{E}[a^{2\,(i)}_{(l-1)}]$$
$$= \sum_i$$
$$= \sum_i Var(w^{(ij)}_{(l)}) \cdot \mathbb{E}[(\tanh(z^{(i)}_{(l-1)}))^2]$$

(with approximation $\mathbb{E}[\tanh(z^{(i)}_{(l)})^2] = \mathbb{E}[(z^{(i)}_{(l)})^2]$)

$$= \sum_i Var(w^{(ij)}_{(l)}) \cdot \mathbb{E}[(z^{(i)}_{(l-1)})^2]$$
$$= Var(w^{(ij)}_{(l)}) \sum_i Var(z^{(i)}_{(l-1)})$$
$$= Var(w^{(ij)}_{(l)}) \cdot n \cdot Var(z^{(i)}_{(l-1)})$$

There fore, we get

$$Var(z^{(i)}_{(l)}) = Var(w^{(i,j)}_{(l)}) \cdot n \cdot Var(z^{(i)}_{(l-1)})$$

$Var(w^{(i,j)}_{(l)}) \cdot n$ muss be equell to 1, because the X ise centerd and whitend and the activation function $\tanh(x) \approx x$ around to 0

There fore, $Var(w^{(i,j)}_{(l)}) \cdot n = 1$

$$Var(w^{(i,j)}_{(l)}) = \frac{1}{n}$$

assumption must be like as

$$W^{(l)}_{ij} \sim N(0, \frac{1}{n})$$

$$b^{(l)}_{i} = 0$$